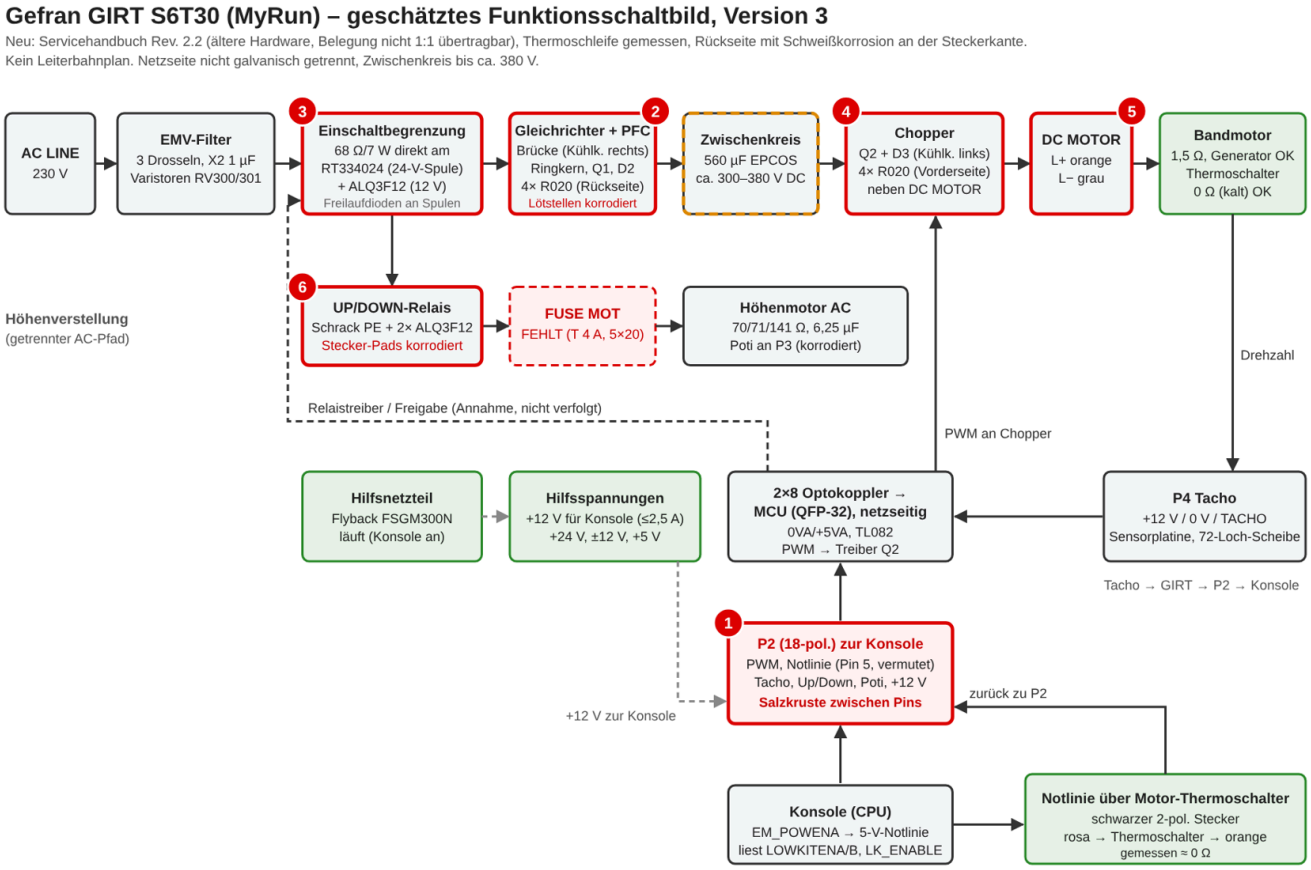
Technogym MyRun – lower kit hardware notes (Gefran GIRT S6T30)

Reverse-engineering notes for the lower kit ("GIRT") of a used Technogym MyRun, collected while repairing one unit. There is **no schematic** for this board. Everything below is either measured on this unit, observed (photos, videos, BLE logs), taken from service manuals of related Technogym drives, or marked as inference.

> **Confidence tags** used throughout: **[M]** measured on this unit · **[O]** observed (photo, video, BLE log) · **[Man]** from a manual of this or a related drive, not verified here · **[I]** inference / hypothesis.
>
> **Safety.** The GIRT is **not galvanically isolated on the mains side**. The DC-link capacitor (EPCOS 560 µF) holds up to ~380 V DC. Measure it (and discharge it through a resistor if above 5 V) before touching the board. Measure live only with leads clamped on while the unit is de-energised. The voltage at *DC MOTOR* rises steeply with belt speed. Never connect anything to the debug header P70 (mains side). Never measure resistance under voltage.

Unit documented: MyRun serial `DCKA2B021008231`, console firmware `WFW0699_BA30.35167`, lower kit Gefran GIRT S6T30, part `0WR01179AG`, board `ECS2458-6` / `ESE6114`, GIRT firmware release 10.05, HW release AG. Name plate: *Gefran Drives and Motion S.r.l., Gerenzano (VA)* (the drives division is now WEG Automation Europe), main power out 0–370 V DC, 1.5 kW **[O]**.

1. Overview



The block diagram is an estimate built from photos of both board sides (German labels); numbers mark the suspicion ranking at the time of drawing, not a verified fault location.

The belt motor is a **brushed DC motor driven by a chopper from the rectified mains** (not an AC inverter drive). The console (CPU board in the handlebar) commands the GIRT over P2: an enable/emergency line, a PWM set-point and tacho feedback. The GIRT's microcontroller (QFP-32, mains side) is separated from the console side by ~10 Vishay VO617A optocouplers **[O]**.

2. Board areas and identified parts

Area	Parts	Function	Tag
Mains input	EMI filter with 3 chokes, X2 1 µF, varistors RV300/RV301	filtering	[O]
Inrush limiter	Schrack RT334024 relay (24 V DC coil, 16 A) + white 7 W 68 Ω resistor	soft start / main relay	[O]
Rectifier / PFC	bridge rectifier (heat sink right), Q1, D2, 4 × R020 shunts on the back	DC link supply, input current sense	[O]/[I]

Area	Parts	Function	Tag
DC link	EPCOS B43630-C9567-M 560 µF	~300–380 V DC	[O]
Motor chopper	Q2 + D3 (heat sink left, markings hidden), 4 × R020 shunts on the front near DC MOTOR	PWM motor drive, current sense	[O]/[I]
Temperature	NTC 10 kΩ on connector P600, screwed next to Q2: 11 kΩ at room temperature, 10.75 kΩ warmed	Q2 temperature	[M]
Incline path	3 × Panasonic ALQ3F12 (12 V, 10 A), Schrack PE014012, **FUSE MOT T4A 5×20**	UP/DOWN relays and fuse for the AC incline motor	[O]
Aux. supply	FSGM300N (DIP-8), transformer ETI1349X4, thermistor RT700	+24 V, +12 V iso, ±12 V, +5 V	[O]
Logic	MCU QFP-32, 2 × 74ACT14, LM2903 (comparator), TL082 (current sense), 24C32 EEPROM, 78L05, 2 × SOT-223 regulators	control, protection, parameters	[O]
Isolation	~10 × VO617A, CPC1017N (OptoMOS)	console ↔ mains-side separation	[O]
LEDs	LINE (blue), +5V (green), STS1 (green), STS2 (red)	see section 7	[O]

Resistor code 4753 on the SMD resistors of the measuring chains means **475 kΩ** (not 47.5 kΩ).

3. Connectors

All connectors sit on the board edge on the solder side of the PCB. The edge collects sweat and is corroded (section 9).

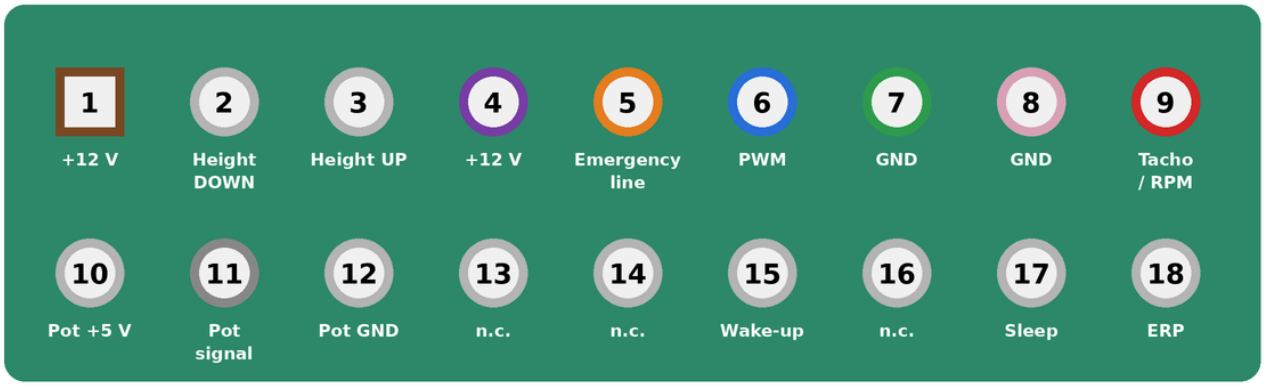
Designator	Purpose	Notes	Tag
P2	18-pin connector to the console	section 3.1	[M]/[Man]
P4	4-pin speed sensor (TACHO / 0V / +12V / 0V)	section 3.2	[M]
DC MOTOR	blade terminals L+ (orange wire) / L− (grey wire) to the belt motor	0 V while the GIRT is locked; 12.4 V DC at 1.2 km/h unloaded (good run)	[M]
UP/DOWN	230 V AC to the incline motor via relays and FUSE MOT	the fuse was missing (glass body gone); replaced with T4A/250 V ceramic, incline works again	[O]
P3	3 pads next to the earth screw	purpose not confirmed; assumed incline potentiometer	[I]
P600	NTC on Q2	10 kΩ NTC	[M]
P70	debug header	**mains side – do not connect**	[O]
thermal loop	black 2-pin inline plug on the motor cable	pink wire (from the P2 cable bundle) → motor thermal switch → orange wire → **P2 pin 5**; ≈0 Ω when cold	[M]
Mains inlet	230 V AC	designator not recorded	–

3.1 P2 – console connector (18 pin, 2 × 9)

GIRT P2 – 18-pin console connector (2 × 9)

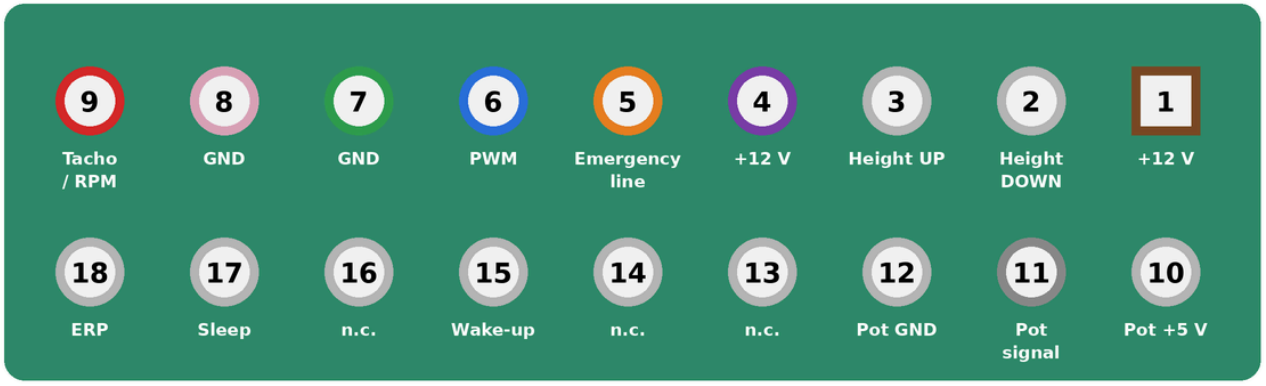
A Plug, cable side (= component side of the header)

Pin 1 top left. Matches the back-probing photo. Wire colours as seen on the cable.



B Board, solder side (back of the PCB)

Mirror image of A: pin 1 (square pad) top right. This is what you see when measuring from behind.



Ring colour = wire colour (where known): 1 brown · 4 violet · 5 orange · 6 blue · 7 green · 8 pink-grey · 9 red · 11 grey; grey ring = not recorded.

Pins 1/4/5/6/7/8/9/11 were measured on this unit. All other names come from the MyRun service manual Rev. 2.2 (older lower-kit generation) and are not verified.

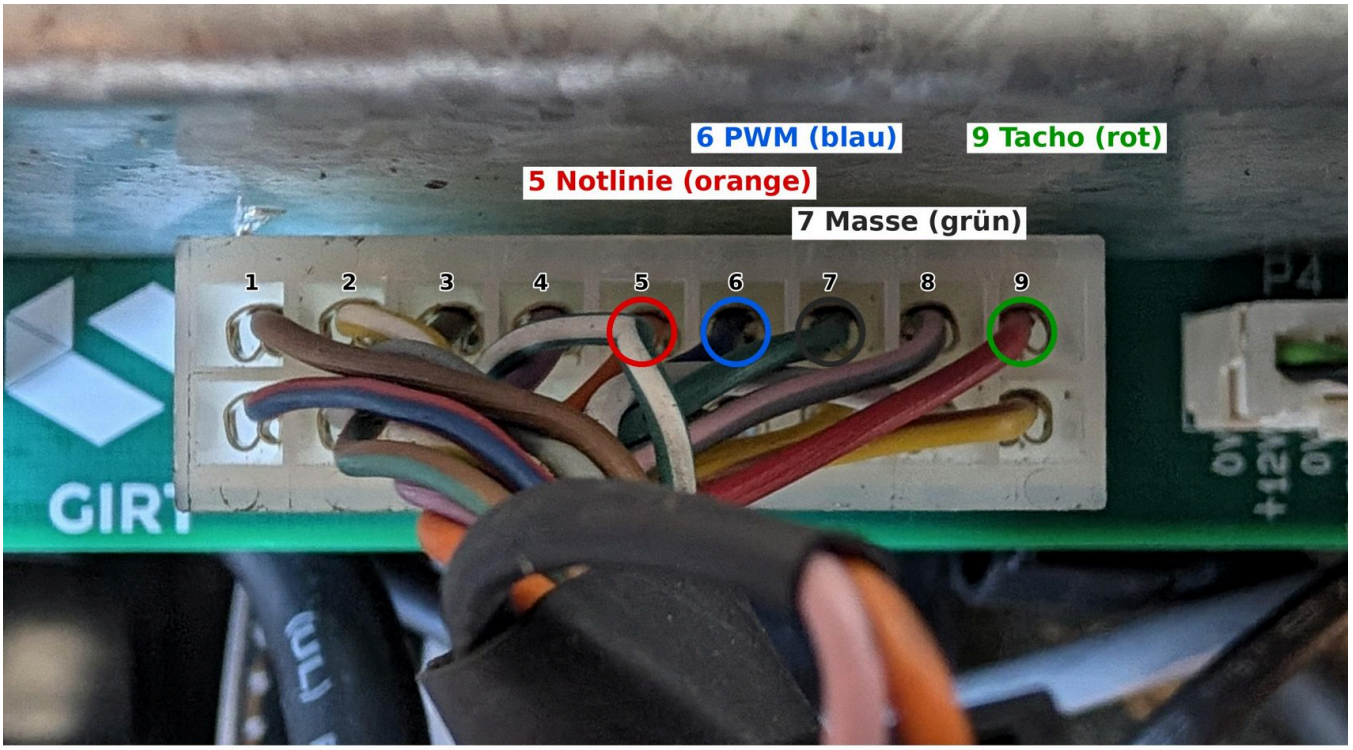
Pins 2/3 switch the incline-motor path – do not probe them while powered.

Pin	Signal	Wire	Observation	Tag
1	+12 V console supply	brown	1 ↔ 4 = 0 Ω	[M]
2	incline DOWN	–		[Man]
3	incline UP	–		[Man]
4	+12 V console supply	violet	1 ↔ 4 = 0 Ω	[M]
5	emergency / enable line	orange	4.55 V at idle and at start; runs through the motor thermal switch	[M]
6	PWM set-point	blue	0 V at idle, ≈0.5 V average during start (console reports 0.077–0.13 %)	[M]
7	GND	green	7 ↔ 8 = 0 Ω; same ground as P4 0V	[M]
8	GND	pink-grey		[M]
9	tacho / RPM	red	≈0.5 V average, changes when the belt moves; 4.3 kΩ to pin 7 (power off); 16 kΩ to P4 TACHO	[M]
10	incline pot +5 V	–		[Man]
11	incline pot signal	grey	≈0.4 V	[M]

Pin	Signal	Wire	Observation	Tag
12	incline pot GND	–		[Man]
13, 14, 16	not used	–		[Man]
15	wake-up	–	expected 0 → ≈3 V → 0 when the belt is switched on (not the mains switch)	[Man]
17	sleep	–		[Man]
18	ERP	–		[Man]

The [Man] names come from the MyRun service manual Rev. 2.2, which describes an older lower-kit generation. The wire colours matched for every pin that was checked.

Orientation. View A of the figure is the plug seen from the cable side (as in the photo below): pin 1 top left, 1–9 on the top row, 10–18 on the bottom row. View B is the board seen from the solder side: the mirror image with pin 1 (square pad) top right. The PCB edge is the bottom row (pins 18…10) in view B.

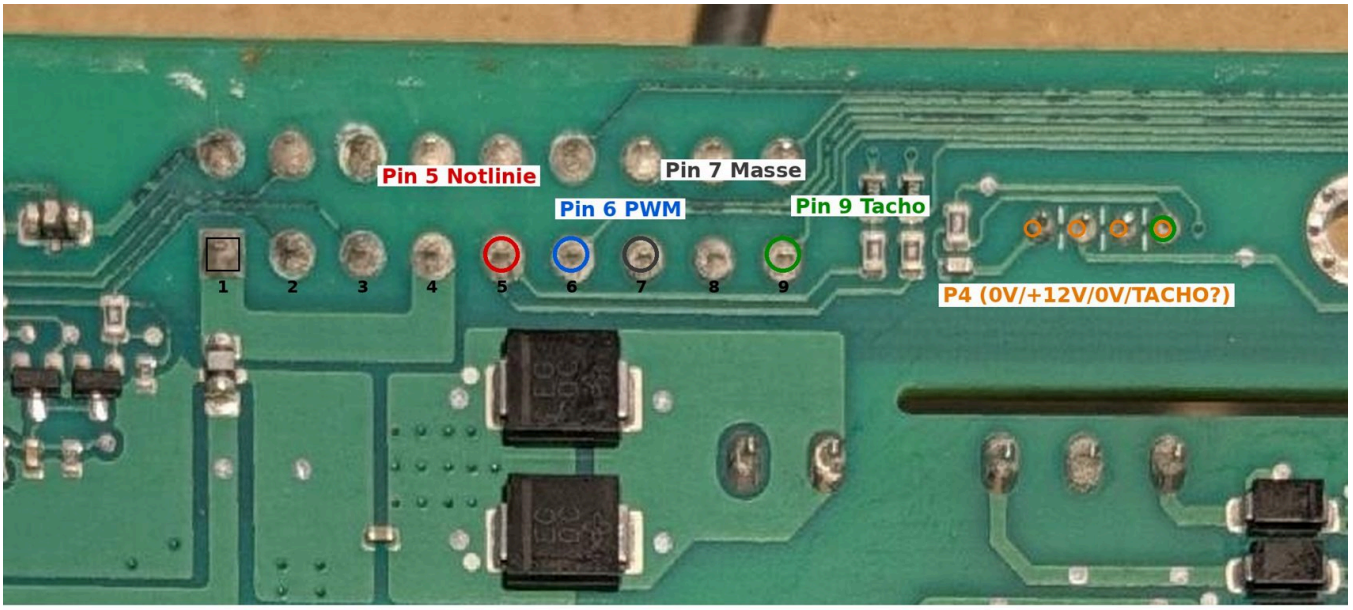


Obere Reihe = Pins 1–9 (Pin 1 links, braun). Schwarz (COM) fest an Pin 7, rot nacheinander:

Pin 5: ca. 5 V solange Freigabe aktiv · Pin 6: einige 100 mV bis wenige V beim Laufen

Pin 9: schwankender Mittelwert beim Laufen, fest 0 V oder ~5 V im Stand · Kontrolle: Pin 1 gegen 7 ≈ 12 V

P2 plug from the cable side with the measurement pins marked (German labels).



P2 innere Reihe (näher zur Mitte), Pin 1 = eckiges Pad links. Messung: Multimeter V DC,

schwarz an Pin 7, rot an Pin 6 (PWM) bzw. Pin 5 (Notlinie) bzw. Pin 9 (Tacho), während des Starts.

P4: Lage aus Fotos abgeleitet. TACHO vermutlich rechts außen (grün) – vorher prüfen:

die beiden 0V-Pins von P4 müssen 0 Ω zu P2 Pin 7 haben.

Solder side of the board: P2 inner row (pins 1–9, pin 1 = square pad at the left in this photo) and the four P4 pads on the right.

3.2 P4 – speed sensor

GIRT P4 – 4-pin speed-sensor connector

Silkscreen: 0V · +12V · 0V · TACHO. Seen from the cable side (plug on); pin 1 = TACHO on the left, as in the photos.

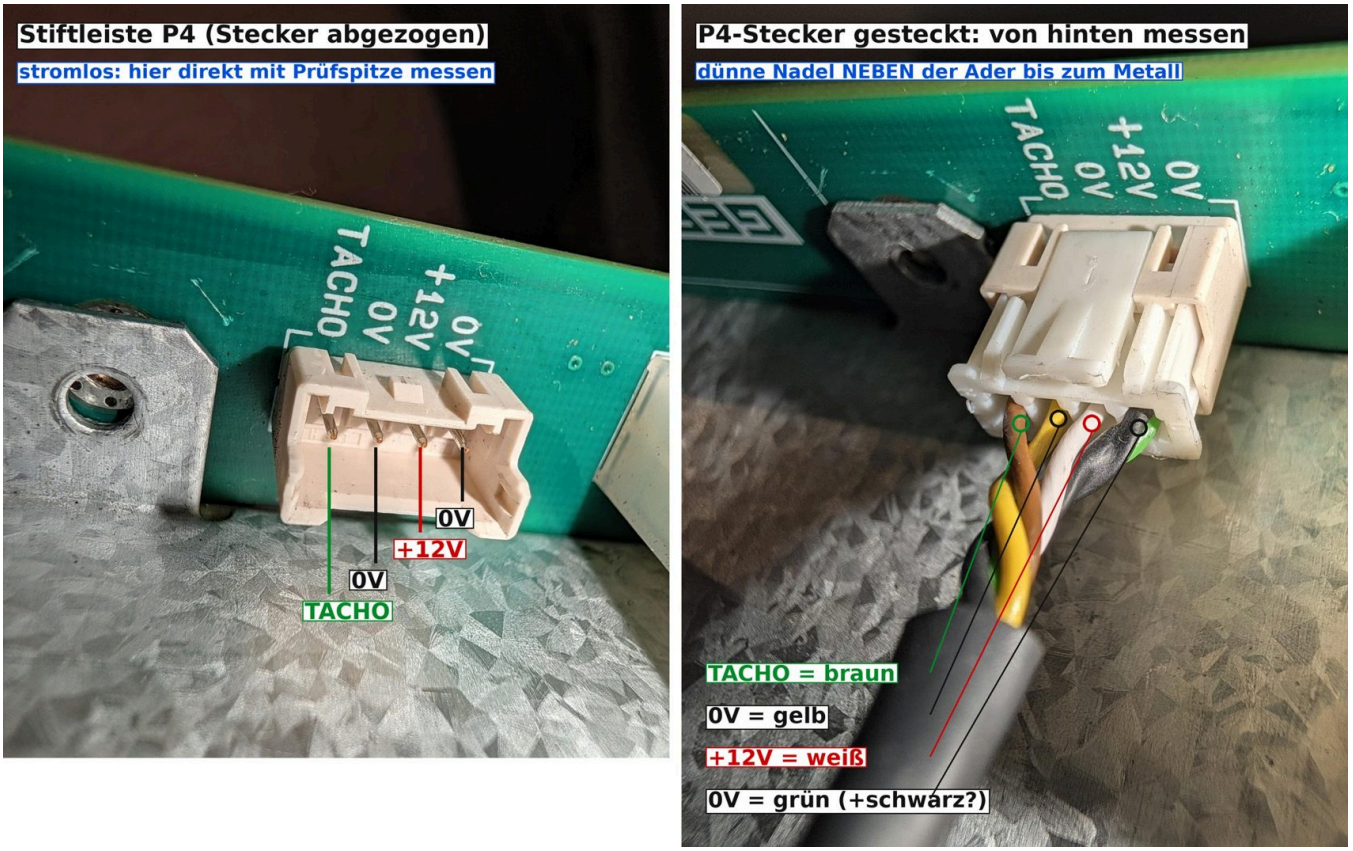
1	2	3	4
TACHO	**0V**	**+12V**	**0V**
brown	yellow	white	green (+ black shield?)

Measured (power off, plugs fitted):

- 0V ↔ P2 pin 7 = 0 Ω (same ground as the console side)
- TACHO ↔ P2 pin 9 = 16 kΩ (a conditioning stage lies between them)
- TACHO ↔ 0V = 12.5 kΩ (pull-up / sensor load) · P2 pin 9 ↔ P2 pin 7 = 4.3 kΩ

Sensor: fork photo-interrupter on the 'INT-SPEED-SENSOR' PCB, slotted disc on the motor shaft.

Wire colours derived from photos; the second 0V wire may be black/shield.



Left: header with the plug removed. Right: plug from behind with the wire colours. German labels.

The sensor is a fork photo-interrupter (marking "1041" / "NXO") on a small PCB labelled `INT-SPEED-SENSOR`, with a finely slotted disc on the motor shaft **[O]**. The encoder frequency reported by the console (`@RENC#`) is about 250 Hz per km/h of belt speed (300–310 Hz at 1.2 km/h) **[M]**. The tacho signal passes a conditioning stage on the GIRT before it reaches P2 pin 9 **[M]**.

4. Console ↔ GIRT signals

`@DUMPIO#` (BLE text channel, see the README) lists the I/O lines between console and lower kit:

Signal	Direction (console view)	Idle / ready	Notes
EM_ENGAGE	out	0	
EM_POWENA	out	1	console asks for power; the next four inputs follow this output
EM_WATCHD	out	1	watchdog
LK_SL_WRK	out	1	
EM_STATUS	in	1	follows EM_POWENA, **not** the GIRT fault state
LK_PW_ENABLE	in	1	
ACC_WK_SENSOR	in	1	dropped to 0 for ≈6 s once while idle
UNFOLD_NC, DSENSOR_B	in	1	
LOWKITENA / LOWKITENB	in	1 / 1	follow EM_POWENA
LK_ENABLE	in	1	follows EM_POWENA
PWM_MOTOR	PWM	0	≈5400–5550 in a good 1.2 km/h run, 8526 when the console pushes harder in a failed start

Other read-only BLE values: `@RPWM#` (PWM %, 0.085 % constant in a good run), `@RENC#` (encoder Hz → km/h), `@STATUS#` (state machine, `MeasSpeed`), `@RERRORLOG#`, `@RSTAT#` (usage), `@RRUNDETMINSP#` (runner-detection threshold, 3.0 km/h on this unit).

The FTMS Treadmill Data speed field (0x2ACD) is the **set-point**, not the measured speed; it stays at the set value while the belt is stuck. Real speed: `@RENC#` / `MeasSpeed`.

5. Fault behaviour (summary)

- Console error codes seen: HF900, HF901, HF902, HF904 (hardware fault after the console loses the speed signal).
- Typical sequence in a failed start **[O]**: START accepted → console raises PWM → encoder stays at 0 Hz, `MeasSpeed` 0 → after ≈7 s EM_POWENA/EM_STATUS/LOWKITENA/B/LK_ENABLE drop (they follow the console output) → console aborts after ≈16 s → `QSHardwareFault` → new HF904.
- The GIRT **does not drive the motor** in that state: 0 V at DC MOTOR (generator voltage is visible when the shaft is turned by hand, so the motor winding is fine).
- The GIRT locks about **1.5 s after START** at the latest and stays locked until it is fully de-energised; the blink code on STS2 (section 7) starts at the same time.
- Good run (17:18, 07.11.): belt ran 95 s at 1.2 km/h, encoder 300–310 Hz, PWM 0.085 %, 12.38–12.41 V DC at DC MOTOR.
- Other values: motor winding 1.5–1.6 Ω over more than 16 rotor positions, thermal switch ≈0 Ω, usage counters from `@RSTAT#`: ≈1390 h powered, ≈342 h belt, ≈3719 km.

Technogym driver codes (same numbering in four drive generations **[Man]**): 01 OH, 02 OC, 03 UU, 04 OU, 05 ST (serial timeout), 06 PE (EEPROM), 07 EdC1 (incline-motor encoder), **08 belt-motor encoder error**, 09 OtM (motor thermal switch), 10 Oli, 11 OLm, 12 OLr. The mapping "STS2 blinks 8 times = code 08 = encoder" is the most likely reading **[I]** but unproven; no other documented board blinks codes on its STS LED.

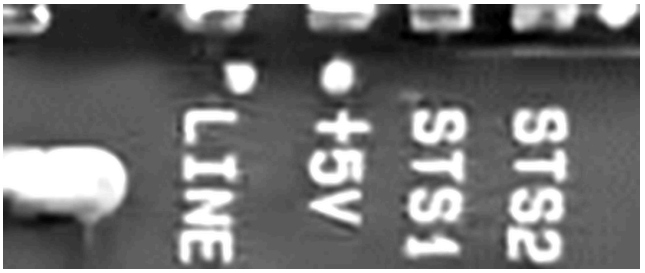
6. Suspected fault area (status of the investigation)

Hypothesis	Rating
Intermittent contact / corrosion in the signal or monitoring path on the GIRT between P2/P4 and the optocoupler/MCU area	most likely [I]
Power stage / drive (Q2, D3, shunts, gate drive)	possible [I]
Motor brushes / commutator	unlikely: 342 h of belt time, but not yet opened and checked
Speed sensor / cable	unlikely: mechanics fine, signal reaches P2 pin 9

Open measurements: the voltage at DC MOTOR (MAX hold) during a fresh failed start after a complete power-off, pad-to-pad continuity of the P2/P4 traces towards the logic area, leakage to neighbouring traces, and a check of the brushes and commutator.

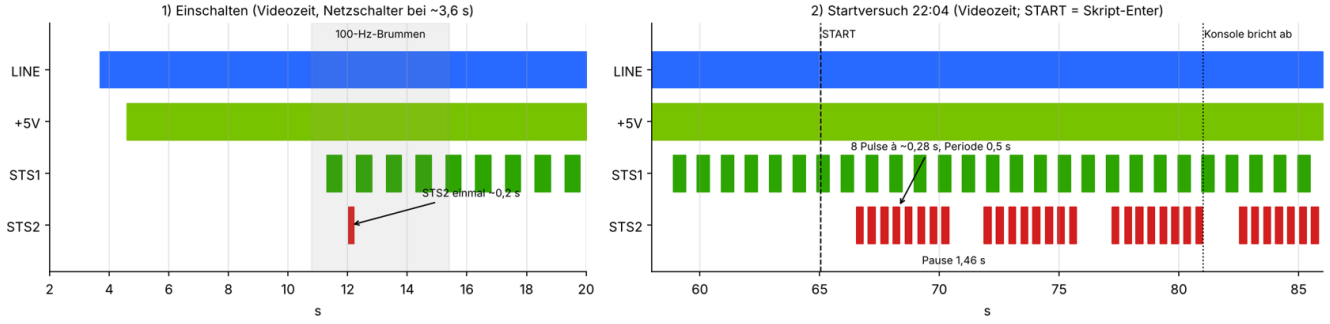
7. LEDs (rear of the board, between D3 and Q2)

Silkscreen: `LINE`, `+5V`, `STS1`, `STS2`.



State	LINE	+5V	STS1	STS2
Boot	on at t0	on at +0.9 s	dark until +7.6 s, then 1 Hz	one 0.2 s flash at +8.3 s
Ready	on	on	1 Hz (0.5 s on / 0.47 s off)	off
Running	on	on	steady on	off
Fault (locked)	on	on	1 Hz	8 pulses (≈0.26–0.31 s on, 0.5 s period), 1.46 s pause, group period 5.33 s

Frame-by-frame timing from phone video **[O]**:

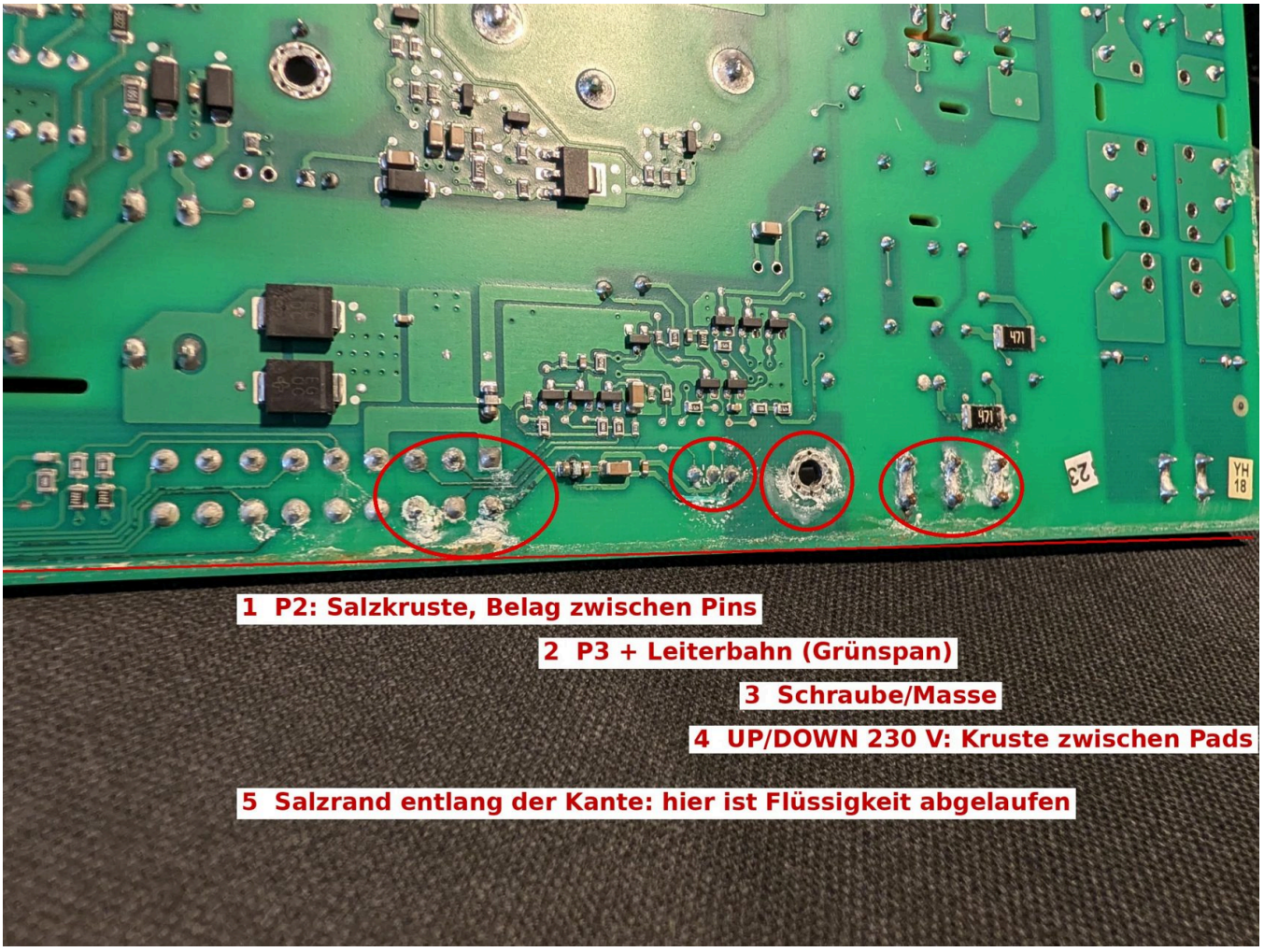


For comparison: on the WEG TPD500 STS1/STS2 are µC status LEDs (boot = asymmetric, normal = symmetric 1 s blink) and carry no error code. The GIRT evidently uses STS2 as its own fault indicator.

8. Repairs and work done on this unit

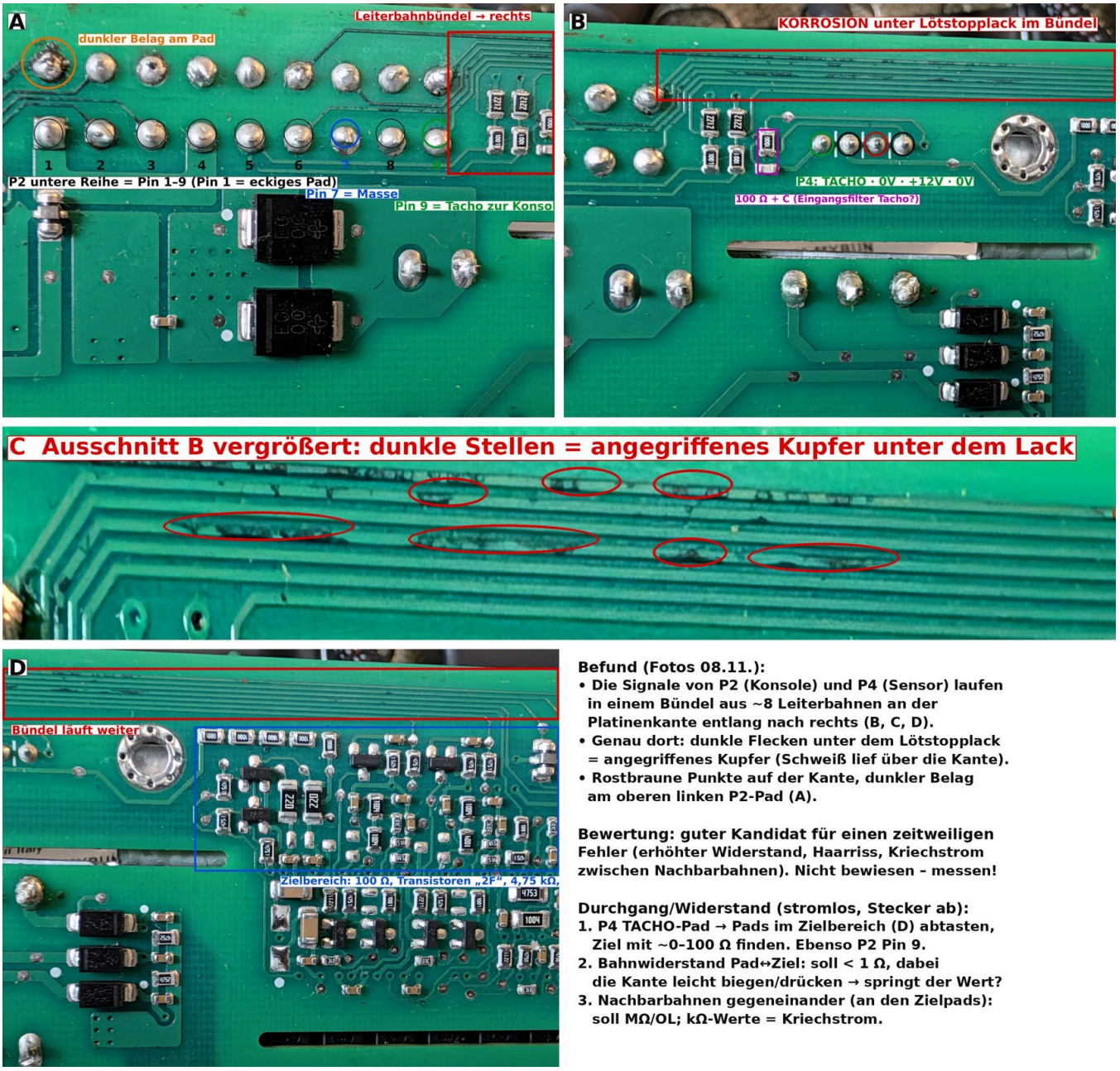
- Incline: missing FUSE MOT replaced (T4A/250 V ceramic, 5×20). Incline test 0 → 2 → 0 % passes.
- Cleaning of salt/sweat residue on the connector edge (isopropanol / water / isopropanol).
- Re-soldered: P2 (all 18 pins), DC MOTOR, UP/DOWN, RT334024 + 68 Ω resistor, bridge rectifier, heat-sink contacts of Q2/D2/D3, all four 4753 resistors with their UJ12 diodes and other suspicious joints. Afterwards the belt ran 95 s once without failing, but the fault came back, so it is not cured.

9. Corrosion findings



Salt crust between the P2 pins, on P3, at the earth screw and between the UP/DOWN pads; a salt line runs along the whole edge (German labels).

The P2/P4 signals run as a bundle of about 8 traces along the top board edge towards the driver/logic area (100 Ω resistors, "2F" transistors, 4.75 kΩ, 22 Ω). Under the solder mask there are dark spots (corroded copper), rust dots at the edge and a dark deposit on one upper-row P2 pad **[O]**. That these traces are interrupted or leaky is a **hypothesis**; it still needs pad-to-pad continuity (≈0 Ω and stable when the edge is pressed) and neighbour-trace checks (OL / MΩ).



Repair idea if a trace is bad: 30 AWG wire-wrap bridge pad to pad, then seal with UV solder mask or protective lacquer.

10. Sources

Service manuals consulted: MyRun Rev. 2.2 (FR, 2016); Technogym Excite Run 500/700/900 Rev. 1.3; Run Personal + Unity Rev. 4.1; Jog Now Excite+ 700; Skillrun Unity 7000 Technical Service Guide Rev. 6.0; WEG TPD500 instruction manual. Their drives are older or different hardware; only the points tagged [Man] are taken over, and the manuals never describe the HF901–HF904 codes or the STS LEDs of this board.